Problems

1 Quiz

1. Let $f(x) = \frac{2000}{\Gamma x+1 \Gamma 2000-x+1} \left(\frac{1}{3}\right)^x \left(\frac{2}{3}\right)^{-x}, x = 0, 1, 2, \ldots 2000$. Find the maximum value for the function.
We know,

$$\begin{aligned}
&\sum f(x)c = 1 \\
&\Rightarrow \sum_{x=0}^{2000} cf \frac{2000}{\Gamma x+1 \Gamma 2000-x+1} \left(\frac{1}{3}\right)^x \left(\frac{2}{3}\right)^{-x} = 1 \\
&\Rightarrow \sum_{x=0}^{2000} \left(\frac{2}{3}\right)^{2000} \frac{2000!}{x!(2000-x)!} \left(\frac{1}{3}\right)^x \left(\frac{2}{3}\right)^{2000-x} = 1 \\
&\therefore x \sim \text{Bin}\left(2000, \frac{1}{3}\right).
\end{aligned}$$

$$\text{Maximum value } = \text{ mode } = (n+1)p = (2000+1)\frac{1}{3}$$

2. If $X \backsim \text{Bin}\left(2021, \frac{1}{2}\right)$, Find the value of $P(X \leqslant 1010) = I_{1/2}(2021-1010, 1010+1) = I_{1/2}(1011, 1011)$.
$\Rightarrow P(X < 1011) = \frac{1}{2}$

3. If $X_1 \sim \text{Bin}\left(5, \frac{1}{2}\right), X_2 \sim \text{Bin}\left(7, \frac{1}{2}\right)$. So, distribution of $(X_1 + X_2) =?$
The condition of independence of X_1 and X_2 are not given, so, reproductive property does not hold.

4. $f(b) = (2020 \times 2019) \times \frac{1}{100} - 404b + b^2 + 202$. Find which value of b for which $f(b)$ maximises.

$$\begin{aligned}
&= E\left(X(X-1) - 2E(X)b + b^2 + E(X)\right. \\
&= E\left(X^2\right) - 2bE(X) + b^2 \\
&= E(X-b)^2 \geqslant E(X-\mu)^2 \quad \therefore \quad 4 = np = 202
\end{aligned}$$

5. If X_1, X_2, X_3, X_4 are distributed an absolutely continuous CDF $F(\cdot)$. What is $P\left(X_1 < X_2 > X_3 < X_4\right)$.
For continuous distribution, $f\left(x_1 = x_2 = x_3 = x_4\right)$ or any $p\left(x_1 = x_2\right) = 0$.

$$\therefore P\left(X_1 < X_2 > X_3 < X_4\right) = \frac{1}{24}$$

6. Expectation of matching problem is always 1 irrespective the value of n. (Remind about the technique of getting expectation as a sum).

7. If $x_i \sim \text{Geo}\left(p_i\right)$ then find $\sum X_i \sim ?$
It can't follow, $\text{NB}(r, \phi)$ as p_i's are not constant. We need equal success probability to get NB from Geometric Distribution.

8. If $X_i \sim \text{Geo}(p_i)$, find the distribution of $Y = \min\{X_i\}$.

$$\begin{aligned}
\min\{X_i\} &= Y \\
F_Y(y) = p[Y \leqslant y] &= 1 - p\left[Y > y\right] \\
&= 1 - p\left[\min X \mid > y'\right] \\
&= 1 - p\left[X_1 > y, X_2 > y_1, \ldots, X_n > y\right] \\
&= 1 - \prod_{i=1}^{n} p\left(X_1 > y\right)
\end{aligned}$$

$$= 1 - \prod_{i=1}^{n} q_i^{y+1}$$

$$= 1 - \prod_{i=1}^{n} \left(1 - p\left(X_i \leq y\right)\right)$$

$$P(Y = y) = F(y) - F(y-1) = 1 - \prod_{i=1}^{n} q_i^{y+1} - 1 + \prod_{i=1}^{n} q_i^{y}$$

$$= \prod_{i=1}^{n} q_i^{y} - \prod_{i=1}^{n} q_i^{y+1}$$

$$= \prod_{i=1}^{n} q_i^{y} \left(1 - \prod_{i=1}^{n} q_i\right) = Q^y(1-Q) = Q^y P$$

9. Show that $E|X| \geqslant |E(X)|$.

$$|E(X)| = \left|\sum_{i=1}^{\infty} x_i \; pi\right| \leqslant \sum_{i=1}^{n} \mid x_i \mid p_i = E \mid X \mid$$

10. If $X \backsim \text{NB}\left(1995, \frac{1}{2}\right)$, find $P(X \leqslant 1994)$.

$$= I_{1/2}(1995, 1995) = -\frac{1}{2}$$

11. Six identical dice thrown independently, S = number of dice showing even face $V(S) = ?$
$\text{Bin}\left(6 \cdot \frac{1}{2}\right)$. $\quad [P(\text{ even face appear }) = P(\text{ odd face appear }) = \frac{1}{2}]$.
$V(S) = npq = 6 \times \frac{1}{2} \times \frac{1}{2} = \frac{3}{2}$.
12. Three independent throw of a dice (fair). It X = number of faces showing 6. Find $E(3-X)^2$.

$$X \sim \text{Bin}\left(3, \frac{1}{6}\right), \quad Y = (3 - X)$$

$$Y \sim \text{Bin}\left(3, \frac{5}{6}\right).$$

$$E\left(Y^2\right) = \frac{2}{3}.$$

13. A box contain 2^n tickets among which $\binom{n}{i}$ tickets are no 1. m tickets are drawn from the box. What is the expected sum of these drawn tickets?

X_i = The no on the ith selected ticket.

$$(1+x)^n = \binom{n}{0} + \binom{n}{1}x^1 + \ldots + \binom{n}{n}x^n$$

$S = X_1 + X_2 + \ldots + X_m = \sum_{i=1}^{n} E\left(X_i\right)$.

$$n(1+x)^{n-1} = \binom{n}{1} + 2^n\binom{n}{2} + \cdots + nx^{n-1}\binom{n}{n}$$

$$E\left(X_i\right) = 0\frac{n_{C_0}}{2^n} + 1 \cdot \frac{n_{C_1}}{2^n} + \cdots + n\frac{n_{C_n}}{2^n} = \frac{1}{2^n}\sum_{i=1}^{n} i \quad n_{C_i} = \frac{n2^{n-1}}{2^n} = \frac{n}{2}$$

14. A box contain a white b black ball. Let c balls are drawn. Find expected value of the no of white balls. (with replacement draw).

(i) $X \sim \text{Bin}\left(c, \frac{a}{a+b}\right)$.

$$E(X) = \frac{ca}{a+b}$$

(ii) Let $X_i = \begin{cases} 1 & \text{white ball drawn at ith draw.} \\ 0 & \text{black ball drawn at ith draw.} \end{cases}$

$$E(S) = E\left(\sum_{i=1}^{c} X_i\right) = \sum_{i=1}^{c} E\left(X_i\right) = \sum_{i=1}^{c}\left(1\frac{a}{a+b} + 0 \cdot \frac{b}{a+b}\right)$$
$$= \frac{ca}{a+b}$$

15. Prove:

$E(\min(X,Y)) \leqslant \text{Min}(E(X), E(Y))$

$E(\max(X,Y)) \geqslant \text{Max}(E(X), E(Y))$

$\text{Min}(X,Y) = \frac{1}{2}[(X+Y) - |X-Y|].$

$\text{Max}(X,Y) = \frac{1}{2}[(X+Y) + |X-Y|].$

$|E(X-Y)| \leqslant E|X-Y|$

$-E|X-Y| \leqslant -|E(X) - E(Y)|$

$\frac{1}{2}\left[-E|X-Y| + E(X) + E(Y)\right] \leqslant \frac{1}{2}[E(X) + E(Y) - E|X| - E(Y)|]$

$\Rightarrow E\,\min(X,y) = \frac{1}{2} = \frac{1}{2}\left[E(X) + E(Y) - E(X-Y)\right] \leq \frac{1}{2}[E(X) + E(Y) - |E(X) - E(Y)|]$

$\Rightarrow \min[E(X), E(Y)]$